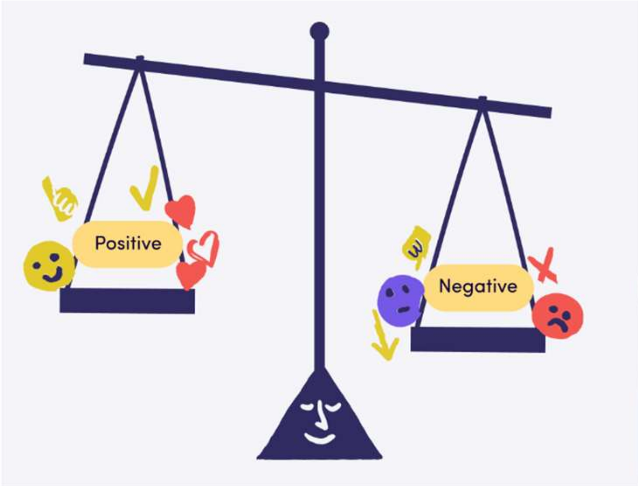
Multi-Agent Reinforcement Learning with Structured Communication Optimization in CityLearn



Interim Report

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Agenda

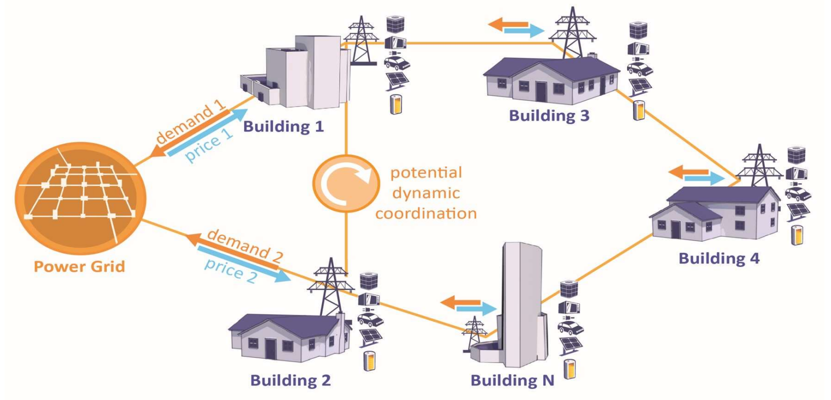
Research Background & Motivation

Motivation:
Growing penetration of renewable energy and electrification requires smart coordination across buildings.
Objective:
Investigate effective MARL structures and communication designs for CityLearn load tracking and analyze their effectiveness.

[1]

CityLearn Simulation Platform

CityLearn

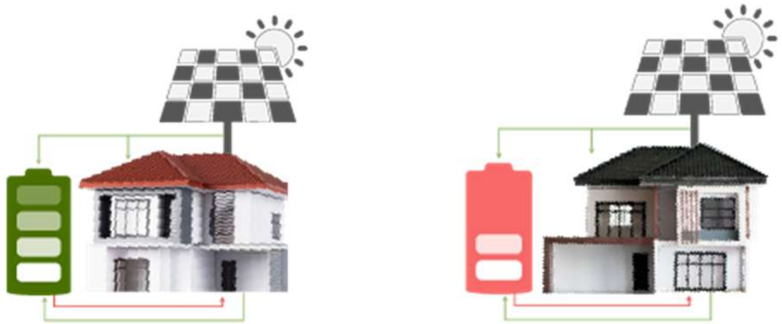


Purpose:
CityLearn provides a standardised open-source environment for demand response and building energy control, enabling reproducible multi-agent reinforcement learning experiments.
Key features:
Simulates communities of buildings with flexible loads, batteries, and HVAC systems; supports continuous state and action spaces for multi-agent control, enabling portfolio-level optimization tasks such as load tracking.
Evolution:
CityLearn v1 introduced an OpenAI Gym-style environment for demand response using deep RL. CityLearn v2 extends the environment with energy-flexible, resilient, occupant-centric and carbon-aware community control, including richer distributed energy resource models and updated evaluation use cases.

Make cities and human settlements inclusive, safe, resilient and sustainable

[2][3]

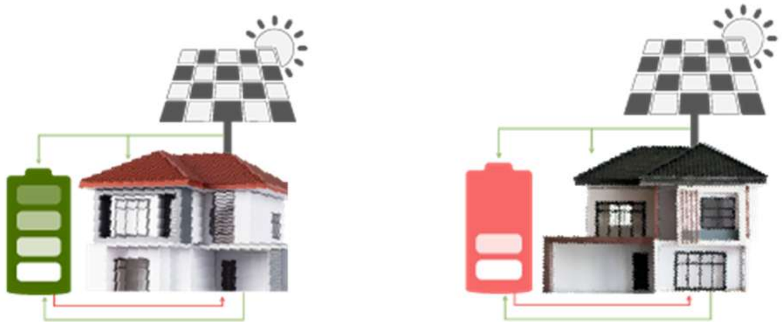
CE1 Benchmark & Evaluation



Scenario:
Annex 96 CE1 defines 25 single-family homes with individual local demand, indoor temperature and solar generation; the portfolio must track a district reference load.
Grouping:
No feeder map or physical topology is provided. Buildings are clustered using flexibility-related features such as battery storage and HVAC capacity.
Train & test:
Vermont(VT) uses January for training and February for testing. Texas(TX) uses August for training and September for testing.
Evaluation metrics:
Primary metric are Normalized Mean Bias Error (NMBE), Coefficient of Variation of RMSE (CV-RMSE) [%] and Thermal Comfort (temperature exceedance hours). Secondary metrics include fairness, cost changes, carbon emissions, changes in total energy consumption, peak demand, peak-to-valley ratio, load factor, and system ramping.

[4][5][6][7]

CE1 Benchmark & Evaluation



NMBE:
Calculated as the mean of the differences between the actual aggregated portfolio load and the reference load profile for each time interval, normalized by the mean reference load and expressed as a percentage. This metric indicates systematic over- or under-consumption relative to the target.

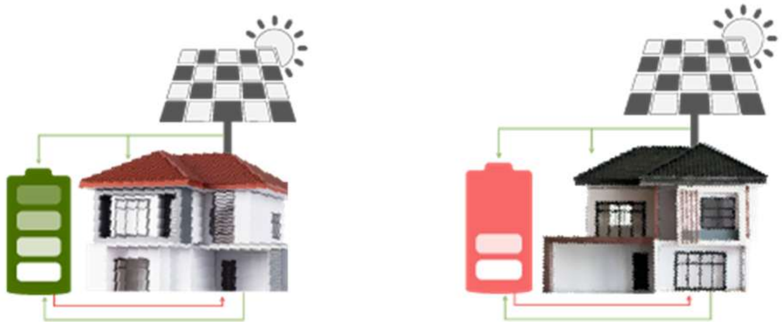
$$\mathrm{NMBE} = \frac{\frac{1}{T}\sum_{t=1}^{T}(y_t - r_t)}{\frac{1}{T}\sum_{t=1}^{T} r_t} \times 100\%.$$

CV-RMSE:
Calculated as the root mean square error between actual aggregated portfolio load and reference load profile, normalized by the mean reference load. This metric captures the overall tracking accuracy including both bias and variance.

$$\mathrm{CV\text{-}RMSE} = \frac{\sqrt{\frac{1}{T}\sum_{t=1}^{T}(y_t - r_t)^2}}{\frac{1}{T}\sum_{t=1}^{T} r_t} \times 100\%.$$

[4][5][6][7]

CE1 Benchmark & Evaluation



Reward function:

$$r_i = -\left[\frac{w_{\text{nmbe}}|\text{NMBE}| + w_{\text{cv\_rmse}}\text{CV-RMSE}}{N} + w_{\text{comfort}}\left(w_{\text{binary}}\,\mathbb{I}_{\text{exceed},i} + w_{\text{degree}}\,d_{\text{exceed},i}\right)\right].$$

The parameter values used in the current code are given:

$$w_{\text{nmbe}} = 1.0, \quad w_{\text{cv\_rmse}} = 1.0, \quad w_{\text{comfort}} = 0.8, \quad w_{\text{binary}} = 1.3, \quad w_{\text{degree}} = 0.3.$$

[4][5][6][7]

Data Split & RBC Baseline

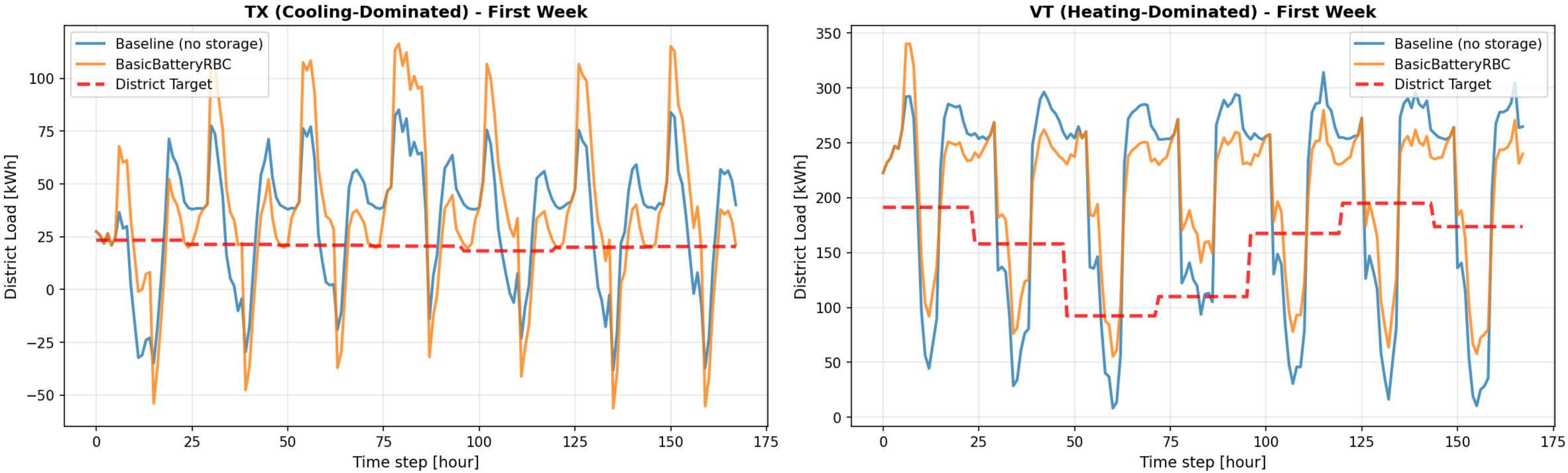
Climate	Training month	Testing month	Status
Vermont	January	February	Fully summarised
Texas	August	September	Comparison future work

Climate	NMBE RBC (%)	NMBE baseline (%)	CV-RMSE RBC (%)	CV-RMSE baseline (%)
TX	89.57	79.12	207.63	163.74
VT	103.93	101.11	129.09	144.56

RBC explanation:
RBC (rule-based control) implements simple if–then logic to charge and discharge batteries and adjust HVAC setpoints based on the current load deviation. It does not learn from data and therefore cannot adapt to dynamic building behaviour, leading to large errors (see the table above).
Observation:
The Annex 96 dataset splits training and testing across months. The RBC rule-based controller uses heuristic charging and HVAC rules to follow the reference load but yields very high CV-RMSE and |NMBE| compared with reinforcement-learning controllers.

[8][9]

Data Split & RBC Baseline

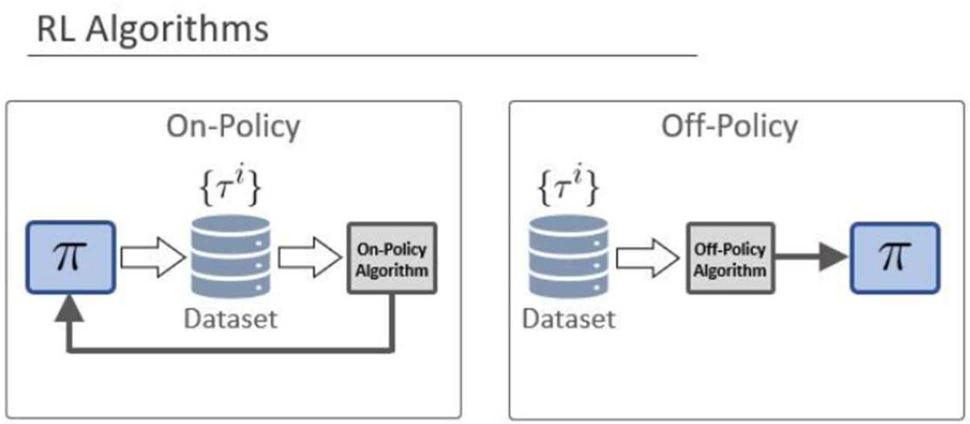


Vermont was studied first because its district target is less flat,

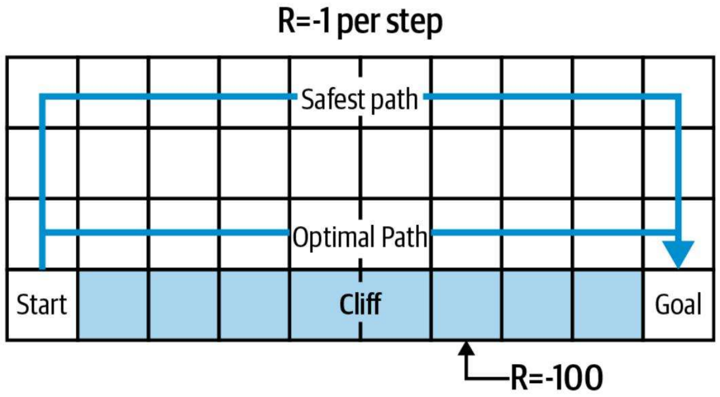
which makes differences between control methods easier to observe during early-stage comparison.

[8][9]

On-policy vs Off-policy Methods



On-policy:
Use trajectories collected with the current policy for updates; stable and easier to control but less sample efficient. Examples: PPO, MAPPO.
Off-policy:
Reuse past experiences via a replay buffer; more sample efficient and exploratory but sensitive to distribution shift between stored data and the current policy. Example: SAC.



[10]

Algorithm Basics: PPO & SAC

Algorithm	**Type**	**Key characteristics**
PPO	On-policy	Clipped policy gradient; stable but sample inefficient; works with discrete and continuous actions
SAC	Off-policy	Soft policy iteration; maximizes entropy-regularized return; uses a replay buffer to improve sample efficiency

On-policy methods: use the latest policy for sampling and update, offering stability but requiring more interactions.
Off-policy methods: reuse past experiences via a replay buffer, providing higher sample efficiency but are susceptible to distribution shift.

[11][12][7]

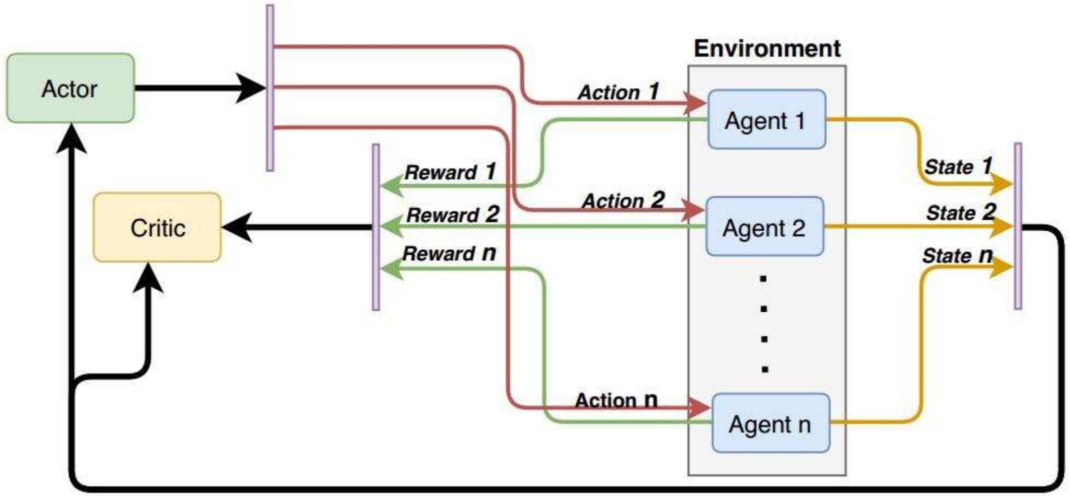
Algorithm Performance: RBC vs PPO vs SAC

Method	Primary rank	CV-RMSE (%)	|NMBE| (%)	Comfort exceedance hour
RBC baseline	15	138.27	109.37	11465
Independent PPO	3	49.76	1.24	2309
Independent SAC	10	46.53	8.93	7470

Observation: The RBC baseline performs poorly with very high CV-RMSE and |NMBE| values and excessive comfort violations; independent PPO dramatically improves tracking and comfort, while SAC improves CV-RMSE but suffers higher |NMBE| than PPO.

[14]

MAPPO & CTDE



Centralized critic:
A shared critic observes the joint state and provide … nt.
Decentralized actors:
Each agent acts using only its local observation; actors can share parameters within groups for efficiency.
MAPPO:
Extends PPO to cooperative MARL with CTDE; employs stable on-policy updates; suits continuous action spaces and cooperative objectives.

MAPPO CODE FROM : marlbenchmark/on-policy: This is the official implementation of Multi-Agent PPO (MAPPO).
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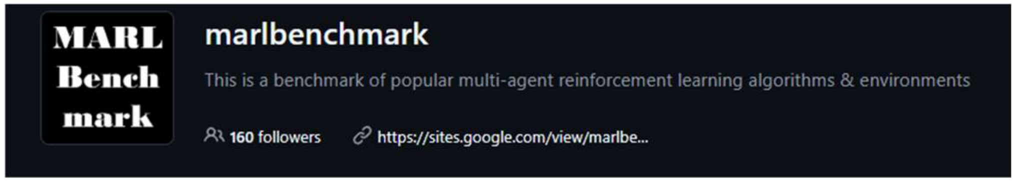


[15][16]

Standard MAPPO vs PPO & SAC

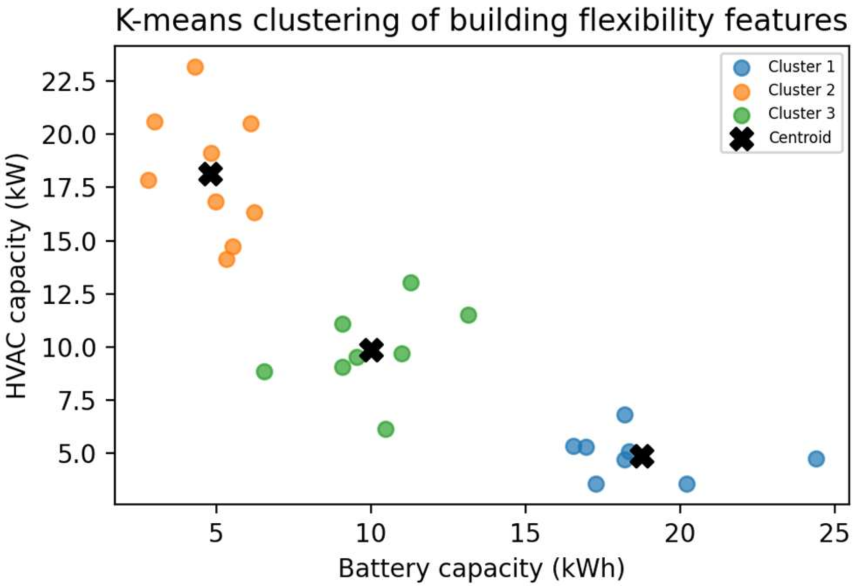
Method	Primary rank	CV-RMSE (%)	|NMBE| (%)	Comfort exceedance hour
Standard MAPPO	12	53.30	5.29	3542
Independent PPO	3	49.76	1.24	2309
Independent SAC	10	46.53	8.93	7470

Observation: Standard MAPPO shows worse CV-RMSE and |NMBE| than independent PPO and higher comfort violations than PPO. While SAC reduces CV-RMSE, its NMBE is higher than PPO and slightly lower than MAPPO; independent PPO delivers the best trade-off among these methods.

[17]

K-means Grouping & Grouped Actor

Selective parameter sharing:
Partition the 25 houses according to flexibility-related features so that agents within a cluster share an actor network.
Clustering method:
K-means clusters buildings using battery storage capacity and aggregate HVAC capacity features.
Grouped actor architecture:
Agents within the same cluster share an actor network, while a central critic oversees all groups; this structure links clustering, selective parameter sharing and building heterogeneity.



[5][6]

Grouped vs Standard MAPPO

Method	Primary rank	CV-RMSE (%)	|NMBE| (%)	Comfort exceedance hour
Grouped MAPPO	6	51.50	2.75	3520
Standard MAPPO	12	53.30	5.29	3542

Observation: Grouping agents via K-means improves performance: grouped MAPPO reduces CV-RMSE and |NMBE| relative to the standard implementation and slightly improves comfort hours, validating the benefits of selective parameter sharing and collective learning.

[5]

Overview of Communication Modules

After standard MAPPO, buildings are divided into groups with K-means.
Buildings in the samegroup share one group actor, but the encoded feature is not changed by any message passing.

$$\tilde{h}_i = h_i.$$

In the communication variants below, the actor first encodes each building observation with an MLP into a hidden feature.
The communication layer then modifies this feature before the action head selects the final action:

$$o_i \rightarrow \text{encoder}_k \rightarrow h_i \rightarrow \text{communication} \rightarrow \tilde{h}_i \rightarrow \text{action head}_k \rightarrow a_i.$$

[18][19][20][21][22][23]

Overview of Communication Modules

Module	Mechanism	Features
CommNet	Average pooling	Compute the mean of messages and update hidden state via a small network
Weighted	Linear weighting	Combine same-group information weighted by α and other-group information weighted by β
PowerNet	Structured topology	Select neighbours based on ring/chain/full topologies; residual updates prevent information washout
GAT	Graph attention	Learn attention weights on a fixed similarity graph to enhance selectivity
TarMAC	Multi-head attention	Learns dynamic weights via a query–key–value mechanism
DIAL	Differentiable messages	Pass continuous messages during training for gradient propagation; approximate discrete messages during evaluation

[18][19][20][21][22][23]

CommNet-Style Mean Communication

The CommNet-style module lets each agent receive the average hidde feature of the other agents:

$$m_i = \frac{1}{|G_k| - 1} \sum_{j \in G_k, j \neq i} h_j.$$

This average message is passed through a small neural network and then added back to the agent's own feature:

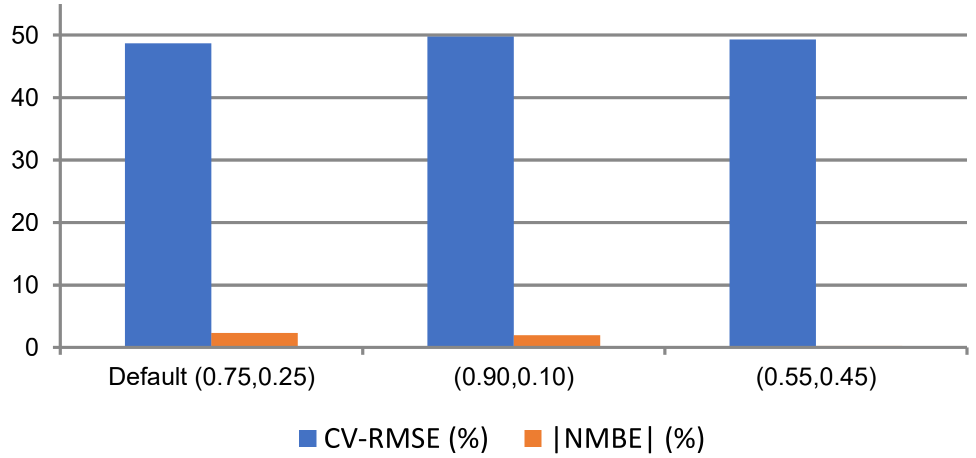
$$\tilde{h}_i = h_i + f_\theta(m_i).$$

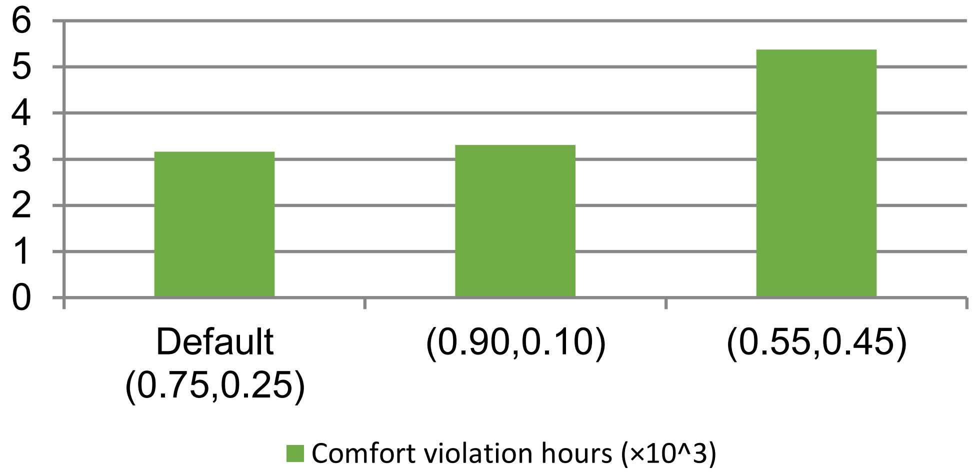
[18][19][20][21][22][23]

Weighted Communication Ablation Analysis

Weighted communication:
Performance depends strongly on same-group and other-group weights. The default setting (0.75, 0.25) achieves the best primary score and comfort but increases peak demand.

Communication weight formula: m_k = α·m_same + β·m_other, where $\alpha > \beta \geq 0$.
Tested combinations include default (0.75,0.25), (0.90,0.10) and (0.55,0.45).





[24][25]

Grouped Communication Variants

Method	Primary rank	CV-RMSE (%)	|NMBE| (%)	Comfort exceedance hour
Weighted default	1	48.68	2.31	3166
Weighted (0.90,0.10)	4	49.75	1.95	3308
Weighted (0.55,0.45)	5	49.28	0.27	5376
Grouped MAPPO	6	51.50	2.75	3520
CommNet	9	48.94	8.11	4873
Comm v2	11	52.92	3.80	4031

Observation: The default weighted controller achieves the best primary performance and lowest NMBE among the grouped variants but increases peak demand; the (0.90,0.10) setting trades off a slight increase in CV-RMSE for lower NMBE, while the (0.55,0.45) setting emphasises cross-group messages and significantly increases comfort violations; CommNet and Comm v2 rely on simple mean pooling and perform worse overall.

[26][27]

Attention Modules

TarMAC:
The TarMAC-style module uses attention. Each agent produces query, key, and value features, and the attention weights are:

$$\alpha_{ij} = \frac{\exp(q_i^\top k_j/\sqrt{d})}{\sum_l \exp(q_i^\top k_l/\sqrt{d})},$$

and the received context is:

$$c_i = \sum_j \alpha_{ij} v_j.$$

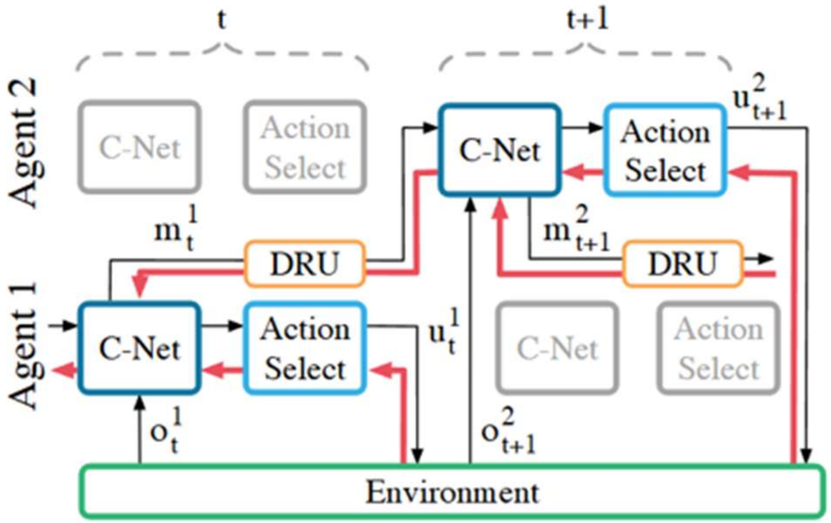
Current implementation still uses mean aggregation and lacks structured filtering. Combining TarMAC with PowerNet could enhance selectivity.

GAT:
The GAT-style module applies masked graph attention on a fixed graph built from the K-means grouping. Buildings within the same K-means group are fully connected, while only a small number of cross-group edges are added between the most similar groups. Each agent then learns attention weights over its allowed neighbors on this predefined graph.

[19][20]

DIAL-Style Differentiable Communication

DIAL:
The DIAL-style module tests a different communication idea. Each agent first turns its hidden feature into a message.
During training, the message remains continuous and differentiable, so gradients can pass through the communication channel.
During evaluation, the message can be discretized.
In this project, the DIAL module is adapted to MAPPO rather than imple mented as the original DQN-based DIAL algorithm.
The purpose is to test whether trainable communication signals help the grouped actors coordinate.



[19][20]

Comprehensive Evaluation & Ranking

Method	Overall rank	Primary rank	CV-RMSE (%)	Comfort exceedance hour	|NMBE| (%)
PowerNet Global	1	2	46.45	4359	0.43
TarMAC	4	7	49.65	3713	3.32
Weighted default	5	1	48.68	3166	2.31
Weighted (0.90,0.10)	7	4	49.75	3308	1.95
Weighted (0.55,0.45)	8	5	49.28	5376	0.27
CommNet	6	9	48.94	4873	8.11
GAT	12	8	51.35	3866	2.92
DIAL	11	13	52.39	6755	7.25

Observation: PowerNet Global performs best overall; the default weighted combination ranks first on the primary metric and has the best comfort but increases peak; TarMAC and (0.90,0.10) have balanced performance; CommNet is relatively weaker.
[10]

PowerNet-Style Structured Information Passing

PowerNet:
The PowerNet-style module tests a more structured way of sharing information. It can use ring, chain, or full topologies to decide which other agents are included in the message. For adjacency matrix A, the selected peer message is:

$$m_i = \frac{1}{\max(1, \sum_j A_{ij})} \sum_j A_{ij} g_\theta(h_j).$$

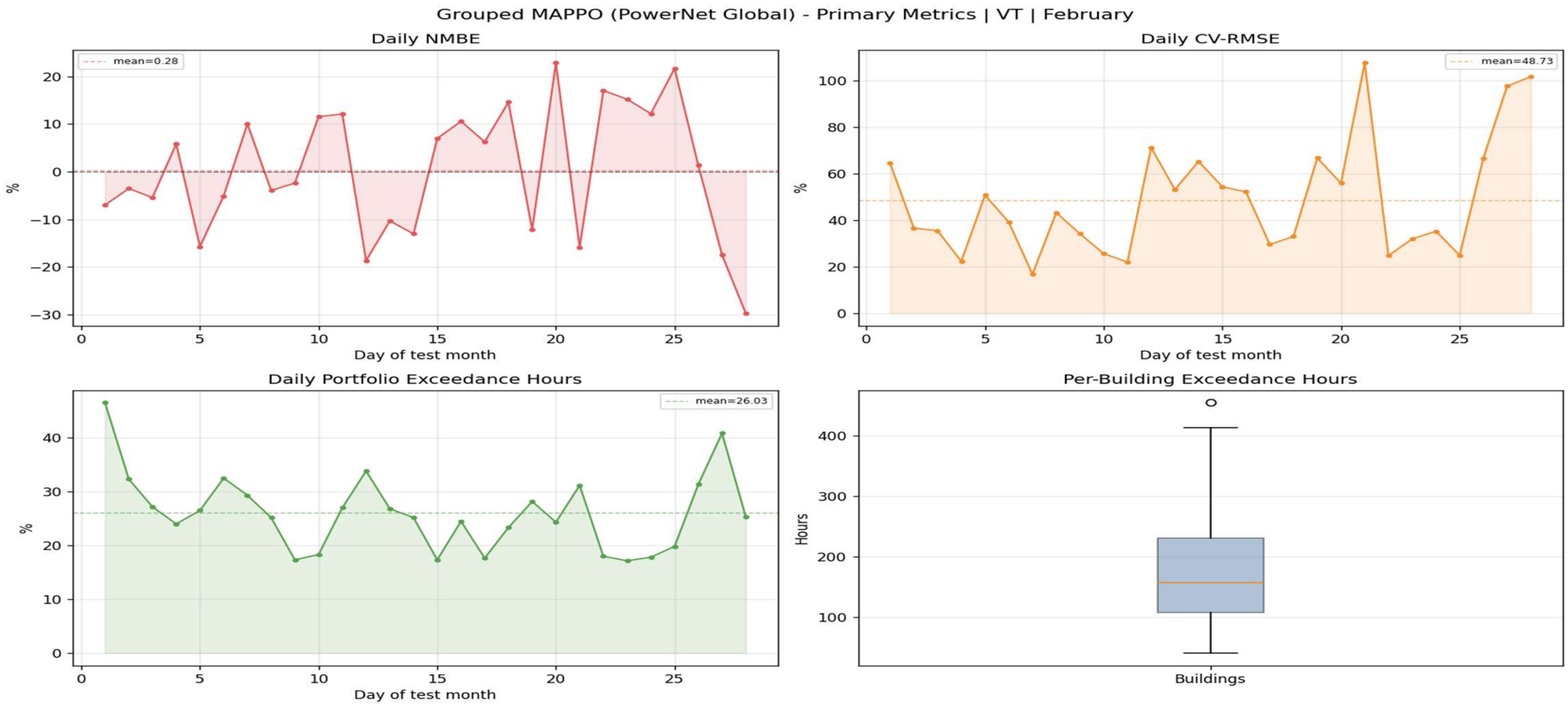
The implementation then combines the agent's own feature with the peer message using a learned residual update. The purpose is to avoid overwriting local information while still allowing agents to coordinate.

[18]

Original Ranking Table

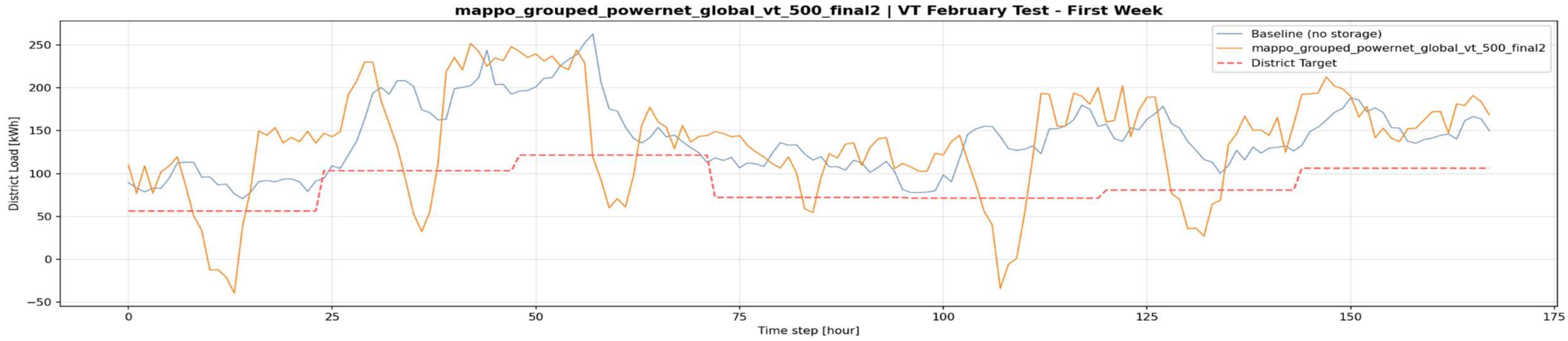
overall_rank	overall_rank_score	primary_rank	experiment	primary_load_cv_rmse_pct	primary_load_nmbe_pct	primary_comfort_exceedance_pct	primary_comfort_exceedance_hours_total
3	5.8704	3	rllib_independent_ppo_vt_80_final2	49.7644	1.2449	13.7440	2309.0
5	6.4815	1	mappo_grouped_comm_weighted_default_vt_500_final2	48.6819	-2.3106	18.8452	3166.0
1	5.1389	2	mappo_grouped_powernet_global_vt_500_final2	46.4515	0.4341	25.9464	4359.0
7	6.9815	4	mappo_grouped_comm_weighted_a090_b010_vt_500_f...	49.7543	-1.9492	19.6905	3308.0
8	8.1481	5	mappo_grouped_comm_weighted_a055_b045_vt_500_f...	49.2801	0.2662	32.0000	5376.0
4	6.2222	7	mappo_grouped_tarmac_vt_500_final2	49.6499	-3.3157	22.1012	3713.0
2	5.6019	10	rllib_sac_vt_500_final2	46.5282	-8.9285	44.4643	7470.0
6	6.7685	9	mappo_grouped_commnet_vt_500_final2	48.9381	-8.1125	29.0060	4873.0
9	8.1944	6	mappo_grouped_vt_500_final3	51.5046	-2.7483	20.9524	3520.0
12	9.2870	8	mappo_grouped_gat_vt_500_final2	51.3497	-2.9221	23.0119	3866.0
11	9.1852	13	mappo_grouped_dial_vt_500_final2	52.3899	-7.2451	40.2083	6755.0
10	9.0278	11	mappo_grouped_comm_v2_vt_500_final2	52.9230	-3.8049	23.9940	4031.0
14	10.3981	12	mappo_standard_vt_500_final3	53.2983	5.2886	21.0833	3542.0
13	9.6389	14	mappo_grouped_powernet_vt_500_final2	50.1563	9.1475	45.7440	7685.0
15	13.0556	15	rbc_baseline_vt_february	138.2660	109.3673	68.3458	11465.0

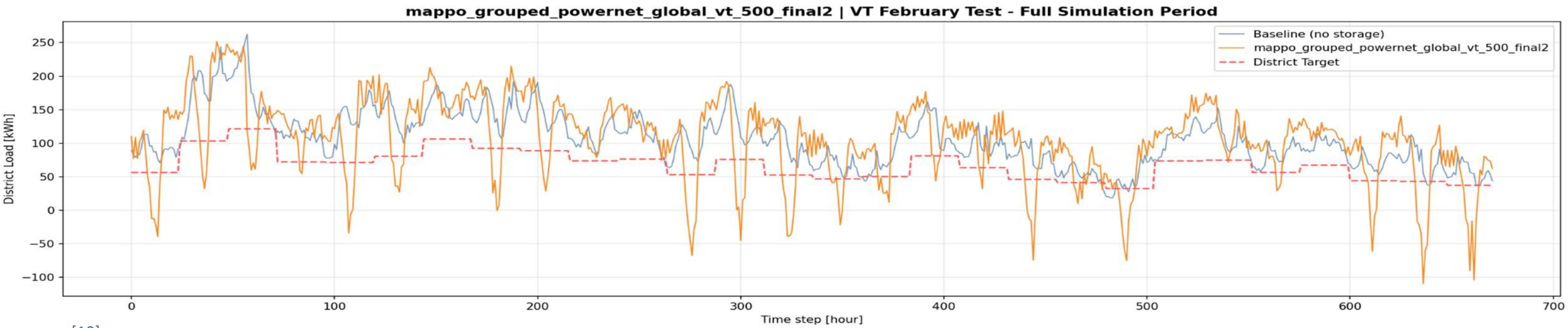
Daily Metrics



[10]

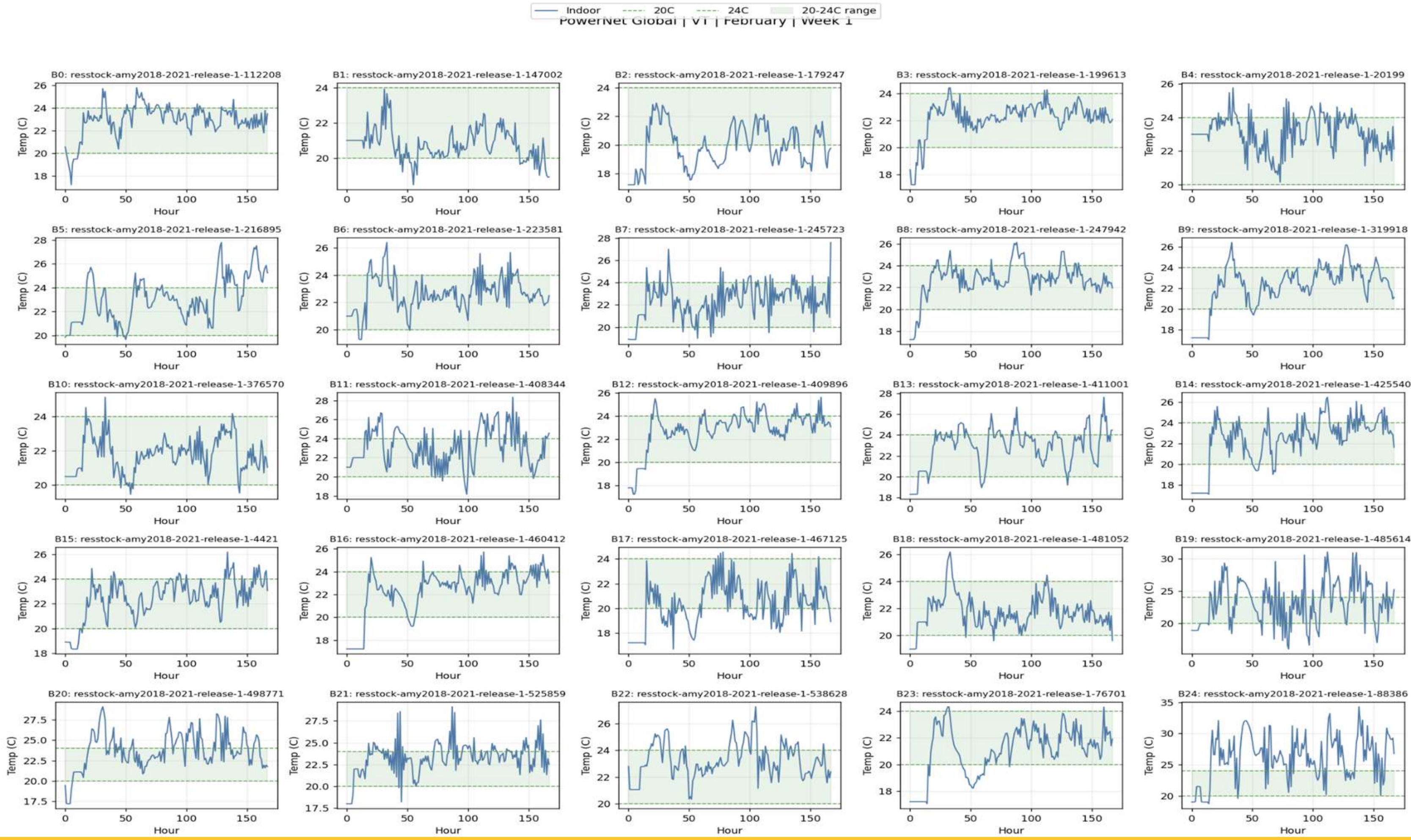
Load Tracking





[10]

Building Temperatures



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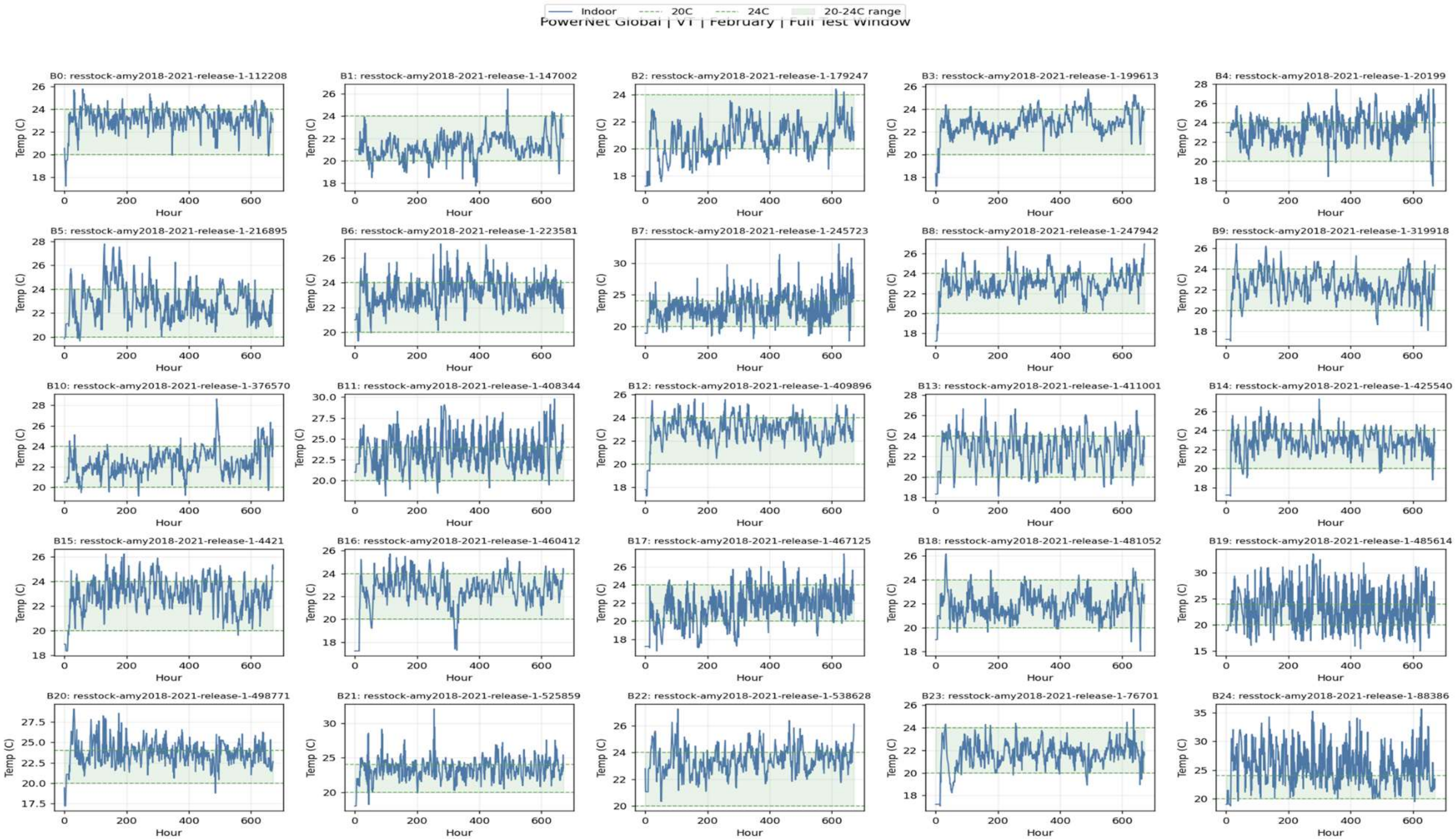
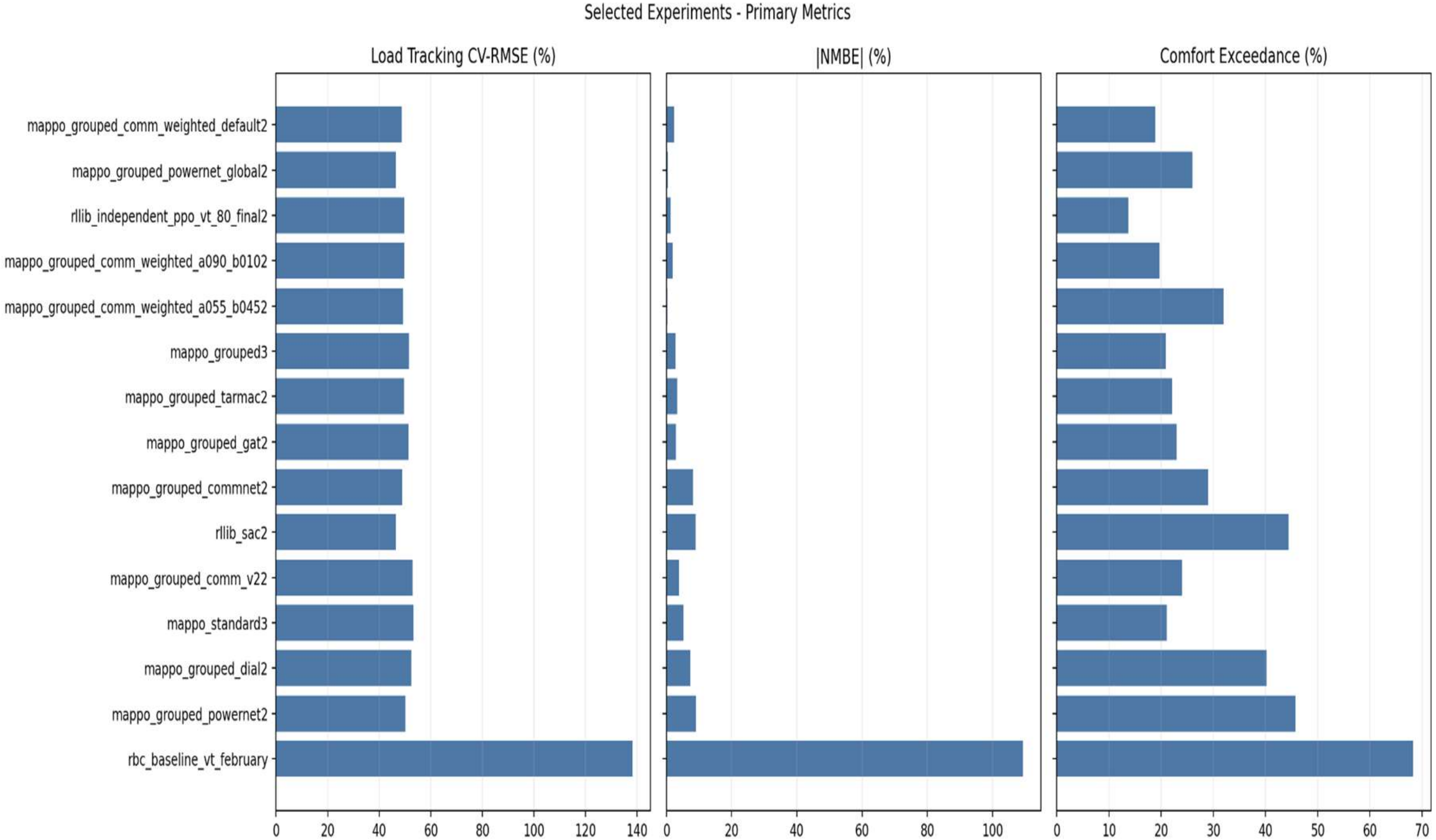
Building Temperatures



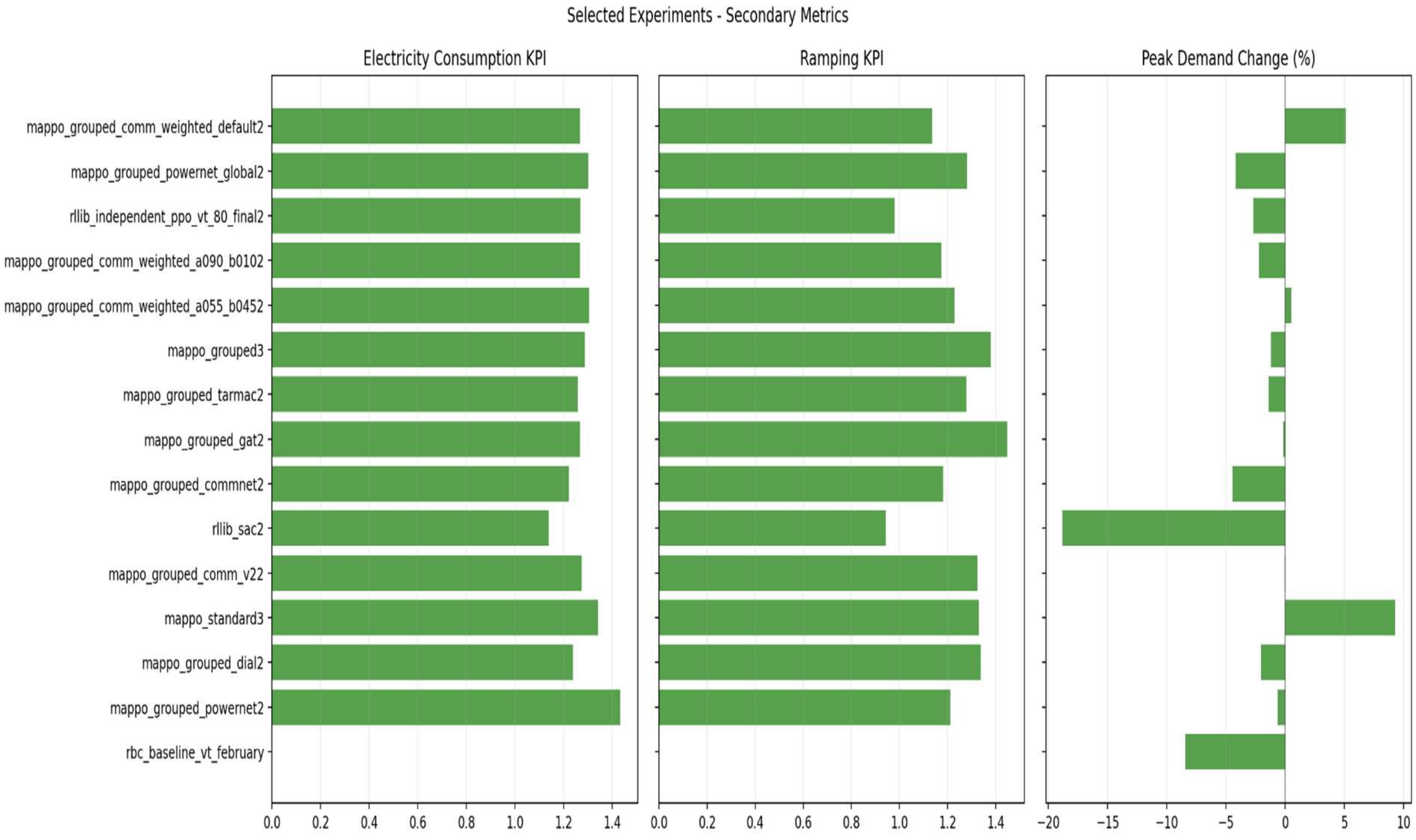
[10]

Primary & Secondary Metrics



[8][10]

Primary & Secondary Metrics



[8][10]

Communication Variant Analysis

PowerNet Global:
Structured information passing retains local features and yields the best overall balance across tracking, comfort and peak metrics.
Weighted communication:
Performance depends strongly on same-group and other-group weights. The default setting (0.75, 0.25) achieves the best primary score and comfort but increases peak demand.
TarMAC:
Learns attention weights to decide whom to listen to; current implementation still uses mean aggregation and lacks structured filtering. Combining TarMAC with PowerNet could enhance selectivity.
DIAL:
Originally designed for discrete actions; in the current continuous-control setting it performs poorly. Weighted and TarMAC variants remain more promising alternatives.
GAT:
The graph attention network weighs messages based on a pre-computed similarity graph derived from building features. This selective scheme moderately outperforms simple mean pooling but relies on fixed adjacency and does not capture dynamic context as effectively as TarMAC or PowerNet.

[18]

Conclusions, Limitations & Future Work

Conclusions:
• Structural choices, such as agent grouping and communication design, significantly influence load tracking, comfort and peak metrics.
• Grouped MAPPO variants outperform rule-based and independent baselines; PowerNet Global offers the best overall balance, while the weighted default controller excels on primary metrics but increases peaks.
• Message weights themselves matter: information from other groups may act as noise or a gain depending on the weight settings.

Limitations:
• Current evaluation is limited to Vermont and single-seed runs.
• The current encoder is a simple MLP and could be upgraded to LSTM.
• PowerNet already outperforms CommNet, but static grouping and current communication designs still limit adaptivity; stronger communication variants, multi-critic structures, and alternatives such as MASAC remain to be explored.

Future work:
• Extend experiments to Texas and perform multi-seed evaluations to assess variability.
• Combine structured filtering (PowerNet) with attention-based weighting (TarMAC) to improve selectivity.
• Test stronger critic designs and new algorithms such as MASAC and value-based methods.
• Enrich reward functions and evaluation metrics by secondary metrics and exploring additional control schemes, then further tune the reward-function weights and continue training the strongest candidate models to enable a fairer comparison.

[24][25][26]

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